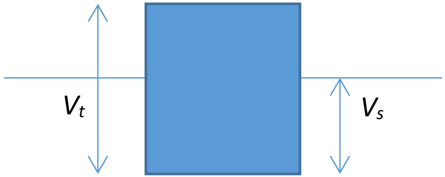
		For option B: $6M - m = -1M + 6m \rightarrow 7M = 7m$ not possible as $M \neq m$ For option C: $6M - m = 1.3M + 8.3m \rightarrow 4.7M = 9.3m$ ok as $M > m$
6	C	Upthrust on ice cube = weight of water displaced  Let the total and submerged volumes of the ice-cube be V_t and V_s respectively. Let also the densities of ice and water be ρ_i and ρ_w respectively. At equilibrium, weight of ice-cube = upthrust which is weight of fluid displaced $$\rho_i V_t g = \rho_w V_s g$$ Hence $V_t = \frac{\rho_w}{\rho_i} V_s$. This equation shows that when V_t of ice melts, it will transform into V_s of water. Hence when the ice cube totally melts, the water level will remain the same.
7	A	By conservation of energy, Initial potential energy = final kinetic energy + work done against friction